Robust Hyperspectral Image Classification Using a Multiscale Transformer With Long- and Short-Distance Spatial–Spectral Cross Attention

Danyang Peng, Haoran Feng, Jun Wu, *Member, IEEE*, Yi Wen, Tingting Han, Yuanyuan Li, Guangyu Yang, and Lei Qu

***Abstract*— Hyperspectral images (HISs) have continuous spectra that can be used to accurately identify the land cover contained within them. Generally speaking, small-scale spectral features can reveal local information within the spectral sequence, while large-scale ones capture global information across spectra. Recently, the proposed Transformer-based hyperspectral image classification (HIC) methods have shown superiority over convolutional neural networks (CNNs) in processing long spectral sequences but struggle with extracting local spectral details. Meanwhile, the multiscale information of the spectral dimension has not been fully explored. To address these issues, we propose a multiscale spatial–spectral Transformer named LSDnet. First, a bilateral filtering-based feature enhancement (BFFE) module is utilized to promote the reliability of shallow features. Then, a long- and short-distance spatial–spectral crossattention (LSDSC) module is proposed to capture both local and global HSI features. Moreover, a multiscale spectral embedding (MSSE) module is introduced to enrich local details among adjacent spectral bands. To enhance the model generalization, the spectral position bias (SPB) coding is designed in the Transformer encoder, adapting to variable spectral numbers of different HSIs. Furthermore, the traditional data partitioning strategy for HIC suffers from information leakage. Therefore, we propose a new data partitioning method to prevent the overlapping between the training and testing data. Meanwhile, an adjustment factor is utilized to balance the number of samples in each category, while a dynamic equilibrium loss (DEL) function is proposed to ensure the training contribution of every category. Experiments on three public datasets validate the rationality and effectiveness of our proposed HIC method. The codes will be available at: https://gitee.com/pdypdy/lsdsc_-tgrs.git.**

***Index Terms*— Hyperspectral image classification (HIC), long- and short-distance attention, multiscale embedding, spatial–spectral cross attention, transformer.**

I. Introduction

With the development of spectral imaging techniques, hyperspectral images (HSIs) have attracted widespread attention due to their spectral properties of electromagnetic spectra. Each pixel of an HSI usually contains hundreds of spectral bands, ranging from visible to near-infrared wavelengths, and corresponding to the spectral reflectance at a specific wavelength [1], [2]. Therefore, compared with other types of remote sensing images, utilizing the spectral properties of HSI for target classification offers greater advantages. Furthermore, HSI has also been widely utilized in the precision agriculture [3], food management [4], and environmental monitoring [5].

In the classification task of HSI, the rich spectral information of HSI not only provides a wealth of classification-related information but also contains redundant information that is disadvantageous to classification. Many researchers have recommended solving this issue by employing dimensionality reduction techniques, such as principal component analysis (PCA) [6], locally linear embedding (LLE) [7], and linear discriminant analysis (LDA) [8]. However, although these methods can reduce the impact of redundant information, they also cause the loss of spectral information. Recently, deep learning tools have been utilized to select spectral information more effectively. For example, Roy et al. [9] proposed an end-to-end unsupervised dual attention reconstruction network that can effectively select bands with better classification accuracy and less redundancy.

Beyond spectral band selection, hyperspectral image classification (HIC) methods are constantly improving alongside the development of machine learning techniques [10]. Traditional machine learning methods typically employ support vector machines (SVMs) [11], multiclass logistic regression, and active learning to construct pixel-based classifiers for analyzing HSI. For instance, Benediktsson et al. [12] utilized various morphological operations to construct spectral–spatial features

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of HSI. Unlike traditional methods, deep learning methods are capable of automatically learning complex features from the data, eliminating the need for manual feature design, which significantly enhances the accuracy and efficiency of classification. In the earliest research attempts, raw spectral vectors were trained directly using unsupervised forms of stacked encoders (SAEs) [13] and deep belief networks (DBNs) [14]. With the development of convolutional neural networks (CNNs), 1DCNNs [15], [16], [17], [18] were employed to extract spectral features, but extracting only spectral information may make different objects display the same spectral feature in HSI. To address this problem, 2DCNN and 3DCNN [19] were leveraged to extract the joint spectral–spatial features of HSI. Furthermore, to add a deeper network structure, Zhong et al. extracted distinguishable spectral–spatial features using spectral and spatial residual blocks separately, achieving more depth and accuracy than 3DCNN [20]. In addition, various other types of deep learning networks were also successfully applied to HIC tasks, for example, the recurrent neural network (RNN), graph convolution network (GCN), and the generative adversarial network (GAN) [21], [22], [23].

Most recently, Transformer networks [24], [25], [26], [27], [28] have also been successfully applied to HIC tasks [29], [30], [31], [32], [33], [34], [35]. The long-distance dependencies among the hundreds of spectral bands have been explored by various Transformer networks. Chen et al. [29] extracted correlation information between long-distance spectral bands using a Transformer. He et al. [30] designed a two-branch network by combining the classical vision Transformer (ViT) [24] and Swin-Transformer [36] models, which can effectively fuse spectral sequence features and spatial features of HSI. Peng et al. [31] proposed a cross-attention spatial–spectral converter (CASST) to achieve weight sharing and reduce the number of parameters in a two-branch network.

Although significant improvements have been made in these Transformer-based HIC methods, the weakness of Transformers in local information extraction prompts researchers to combine local feature extractors, such as CNNs, with Transformers to form more effective HIC methods. He et al. [32] proposed combining a Transformer with a visual geometry group (VGGnet) to extract spectral information and enhance interband correlation. Sun et al. [33] suggested using Gaussian functions in combination with spectral features, resulting in integrated global–local spatial features. Wang et al. [37] utilized an information complementarity and sequential perspective for cross-attention feature fusion based on Transformers, allowing for the effective mining and integration of enhanced features at various levels. Gao et al. [38] developed S2PNet, integrating Transformers to mine the cross-representation between spatial distributions and spectra in remote sensing images. Mei et al. [34] proposed a model combining grouped pixel embedding with a Transformer. The grouped spectral embedding module proposed by Hong et al. [35] obtains the correlation between neighboring spectral bands to enhance the extraction of local information. In these methods, the extraction of local features is mainly realized by CNNs, with a focus on extracting spatial local information. However, in the spectral dimension, an efficient global–local feature extraction method is still lacking.

On the other hand, multiscale features have been proven to be important for HIC. Meng et al. [39] extracted multiscale features by using convolutional kernels of different sizes. Shi et al. [40] proposed a channel shrink soft split module to extract the missing local information in large-scale tokens and performed dimensionality reduction on channels at the same time. Once again, those multiscale features were mainly focused on the spatial dimension, they could only extract single-scale spectral information and could not fully describe the correlation between spectral bands.

Furthermore, limited by the difficulty of manual annotation, there are few labeled HSI datasets available. Most deep-learning-based HIC methods prepare training data by taking patches according to the central pixel, which has the following problems. First, these training HSI patches may contain pixels that are not consistent with the category of the center pixel, and the features extracted from these inconsistent pixels will interfere with the classification results. Second, the method of taking patches centered on a pixel may cause information leakage between training data and testing data, which seriously affects the robustness of HIC methods [41], [42].

In this article, we propose a multiscale spatial–spectral Transformer network for HIC named LSDnet, based on our designed long- and short-distance cross-attention and multiscale spectral embedding (MSSE) strategies. First, the shallow features of HSI images are obtained through 3-D convolution. Then, these shallow features are further enhanced by bilateral filtering to highlight effective features and preserve edge information. To promote information interaction between different ranges of spectral bands, we propose an MSSE module to obtain multiscale feature maps. Finally, these feature maps are grouped into spectral series and imported into the long- and short-distance cross-attention module to obtain effective classification features with the ability to describe both long- and short-range spectral information. The contributions of this article are summarized as follows.

1) We have proposed a long- and short-distance spatial–spectral cross-attention (LSDSC) module to alleviate the issue of the insufficient ability of classical ViTs to describe short-distance spectral features.
2) We have designed an MSSE module to achieve the fusion of spectral information at different scales, enhancing the correlation of neighboring spectral bands.
3) We have proposed a bilateral filtering-based feature enhancement (BFFE) module to enhance the spatial features of central-consistent pixels and suppress the features of central-inconsistent pixels.
4) We have proposed a new Transformer-based HIC network by leveraging the proposed BFFE module, the MSSE module, and the LSDSC module, and the competing classification results are obtained on three public datasets.

II. Proposed Method

The architecture of the proposed LSDnet is displayed in Fig. 1. We have conducted four procedures to extract effective classification features from HSI, ranging from shallow-to-deep levels. These include shallow feature extraction (SFE), BFFE, MSSE, and LSDSC.

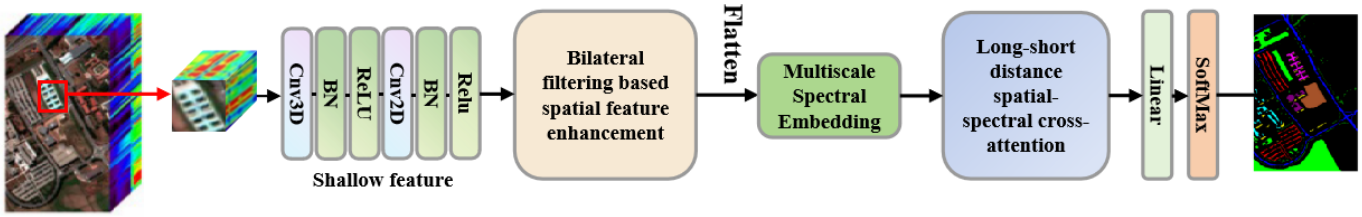


Fig. 1. Architecture of LSDnet.

A. Shallow Feature Extraction

Shallow spatial features and spatial–spectral features typically represent the low-level visual characteristics of HSI, such as spatial edge information and local spatial–spectral correlation. Convolutional operations have a strong ability to extract these shallow and local features. Therefore, we propose an efficient and simple CNN-based module for extracting shallow features from HSI.

Specifically, for the original HSI data $Z \in R^{H\times W\times L}$, where $H \times W$ represents the spatial size and L is the length of the spectral sequence, each pixel has L spectral bands. We first divide Z into small patches $P \in R^{t\times t\times L}$, where $t \times t$ indicates the spatial size. The central pixel of each patch is denoted by (X_i, X_j), where $0 \le i \le m$ and $0 \le j \le n$, and the category of each patch is determined by the label of the central pixel. Subsequently, a 3-D convolutional layer and a 2-D convolutional layer are concatenated to form the SFE module, which is implemented at the beginning of the proposed LSDnet.

B. Bilateral Filtering-Based Feature Enhancement

Most existing HIC methods utilize the label of the central pixel to indicate the category of an HSI patch. Then, the local information within the image patch is employed to predict the category of the central pixel during the inference stage. However, when some pixels in an image patch are inconsistent with the category of the central pixel, the image information of these inconsistent pixels reduces the reliability of classification features. For example, in Fig. 2(a), the central pixel belongs to category 2, but most existing methods categorize all pixels in the image patch into category 2 and assign all pixels equal weight, even if some pixels actually belong to categories 1 or 3. Therefore, to improve the reliability and effectiveness of shallow features, we need to promote the features of central-consistent pixels and suppress the influence of central-inconsistent features. As shown in Fig. 2(b), we propose using the central pixel as a criterion to classify all pixels within the image patch. Pixels with high similarity to the central pixel are labeled as central-consistent pixels, and the remaining pixels are labeled as central-inconsistent pixels.

Liao et al. [43] proposed a Gaussian-weighted feature tokenizer for feature extraction, where pixels farther away from the center are assigned smaller weights. However, they only considered the positional relationship between the central pixel and the neighboring pixels but neglected their similarity. We propose leveraging the bilateral filter (BF) to solve this problem. The bilateral filter can fully reflect the positional relationship between the central pixel and the neighboring pixels and, at the same time, effectively differentiate between central-consistent pixels and central-inconsistent pixels among the neighboring pixels. In addition, the BF can preserve more edge features than the Gaussian filter (GF) [44].

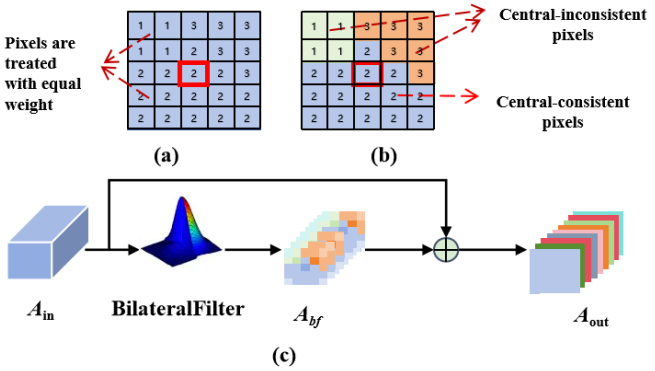


Fig. 2. (a) Equal weight in HSI patch. (b) Category related weight in HSI patch. (c)BFFE module.

The proposed BFFE module is shown in Fig. 2(c). The feature map $A_{\text{in}} \in R^{t'\times t'\times L'}$ represents the shallow feature maps extracted from the SFE module. We redistribute the distribution of A_{in} through a BF function to obtain the feature-enhanced map A_{bf}, which highlights the central-consistent features and suppresses the central-inconsistent features. However, A_{bf} ignores the original features, so we add the original input A_{in} to A_{bf} to obtain the final enhanced shallow feature map A_{out}. The above operations can be expressed as follows:

$$A_{\text{out}} = \text{BF}(A_{\text{in}}) + A_{\text{in}} \tag{1}$$

where $\text{BF}(\cdot)$ denotes the bilateral filter function and $+$ denotes pixel-by-pixel summation.

In particular, the BF is calculated as follows:

$$\begin{aligned}\text{BF}(i,j,k,l) &= g_s(x_w - x)\cdot f_r(I(x_w) - I(x)) \\ &= \exp\left(-\frac{(i-k)^2+(j-l)^2}{(2\sigma_d^2)} - \frac{I(i,j)-I(k,l)}{(2\sigma_r^2)}\right)\end{aligned} \tag{2}$$

where (k,l) denotes the central pixel position, (i,j) denotes the neighboring pixel position, and $I(i,j)$ and $I(k,l)$ denote the pixel values at pixels (i,j) and (k,l), respectively. Moreover, σ_d is the spatial domain parameter and σ_r is the pixel

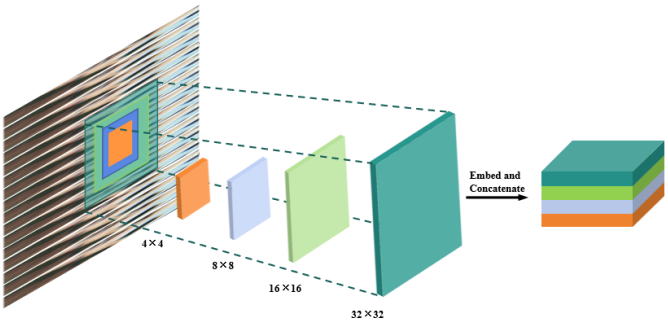


Fig. 3. Structure of MSSE module.

magnitude domain parameter. This weight matrix represents both the spatial location and similarity between the central pixel and the neighboring pixels. The central-consistent pixels closer to the center pixel have higher spatial weights, and the central-inconsistent pixels relatively far from the center pixel have lower spatial weights.

C. Multiscale Spectral Embedding

The grouped spectral embedding proposed in [35] can effectively acquire local spectral features, but it only enables the extraction of single-scale spectral information and cannot describe the correlation between spectral bands at different scales. Recently, the combination of multiscale features with Transformer has shown significant results in visual tasks such as image classification. Fan et al. [45] created a multiscale feature pyramid, where the shallow layer provides low-level visual information with high spatial resolution but simple features, while the deep layer provides spatially coarse yet complex high-dimensional features. Chen et al. [46] designed a multiscale feature fusion method for a two-branch Transformer that uses cross attention to fuse small and large image patch features extracted from the two branches, respectively. Furthermore, Guo et al. [47] proposed utilizing multiscale local Transformations to model frames within local windows in long videos, where visual features, optical flow and extracted features are organized into a feature pyramid and then fused by a fusion layer. In this regard, we design an MSSE module to achieve feature fusion of spectral bands at different scales, enhancing the correlation of local spectral bands.

The details of the MSSE module are shown in Fig. 3. For the final enhanced shallow feature map $A_{\text{out}} \in R^{t' \times t' \times L'}$, we flatten it and then splice it to obtain a new feature map $X \in R^{(t' * t') \times L'}$. We learn a function $F_{k \times k}(\cdot)$ that maps X to a new token $F_{k \times k}(X)$ with channel size c, where $F(\cdot)$ is a 2-D convolutional operation with a kernel size of $k \times k$, step s, and padding p. The output 2-D token map $F_{k \times k}(X) \in R^{M \times N \times c}$ has the following height and width:

$$M = \lfloor \frac{t' * t' - k + 2p}{s+1} \rfloor, N = \lfloor \frac{L' - k + 2p}{s+1} \rfloor. \quad (3)$$

Specifically, the enhanced feature map X is sampled using four convolutional kernels of different sizes, and four embedding layers of the same size are obtained by adjusting the step size and padding. In practice, we use four convolutions of different sizes: 4×4, 8×8, 16×16, and 32×32. Thus, given a feature $X \in R^{(t' * t') \times L'}$, we can obtain four features: $F_{4\times4}(X)$, $F_{8\times8}(X)$, $F_{16\times16}(X)$, and $F_{32\times32}(X)$, all with the same size $M \times N \times c$. The four feature layers with different scales are concatenated to obtain an MSSE layer $T \in R^{M \times N \times 4c}$. As shown in Fig. 3, the four cubes represent four different scales of feature maps.

D. Long- and Short-Distance Spatial–Spectral Cross Attention

Although the Transformer is very effective in dealing with long correlations between HSI spectral bands, it cannot represent local variations between adjacent spectral bands and cannot extract multiscale spectral information [35]. To address these issues, we design a long- and short-distance cross-attention mechanism based on MSSE to achieve the fusion of spectral information at different scales and preserve the local and global features of HSI. The structure of the LSDSC module is shown in Fig. 4, which consists of two parts: the short-distance spectral embedding and the long-distance spectral embedding.

1) Short-Distance Spectral Embedding: HSI has hundreds of spectral bands and produces continuous spectral features with fine disparities. Therefore, capturing the local variations between spectra is a key factor in the HIC task. Most existing Transformer-based methods use the classical ViT to deal with the sequential information between spectra of HSI, but lose the ability to capture the subtle differences in spectra [35]. Therefore, we propose a short-distance spectral embedding to learn spectral representation from adjacent spectra and capture subtle differences between them. First, the embedding with multiscale spectral information is input into the short-range spectral embedding. Then, the adjacent T spectral bands are embedded and grouped. We take $T = 3$ as an example in Fig. 4, and the local variation between neighboring spectral bands is obtained by using self-attention within each group. The short-distance grouping can be formulated as follows:

$$B_s = (B_1, B_2, \ldots, B_{\text{T}}), \ldots, \left(B_q, B_{q+1}, \ldots, B_{q+\text{T}}\right), \ldots, (B_{N-T}, B_{N-T+1}, \ldots, B_N) \quad (4)$$

where B_s denotes the short-distance spectral group, while $(B_q, B_{q+1}, \ldots, B_{q+T})$ represents one of the short-distance spectral groups. N denotes the number of spectral bands, B_q denotes the qth spectral band of the HSI patch, and T denotes the number of spectral bands in each group.

2) Long-Distance Spectral Embedding: When computing attention, it is challenging to establish dependency correlations between two spectral bands that are far apart without the assistance of large-scale spectral information. However, in this article, with the help of our proposed MSSE module, sufficient contextual information are provided for the spectral bands that are far apart, making the long-distance multiscale spectral information more effective and meaningful.

First, the spectral bands are sampled at fixed intervals as illustrated in Fig. 4. In this figure, the green-marked spectral bands constitute one group, the yellow-marked spectral bands form another group, and the remaining spectral bands are grouped similarly. Self-attention is applied to each group to learn spectral representations from long-distance spectra and capture the long-range correlations between spectra.

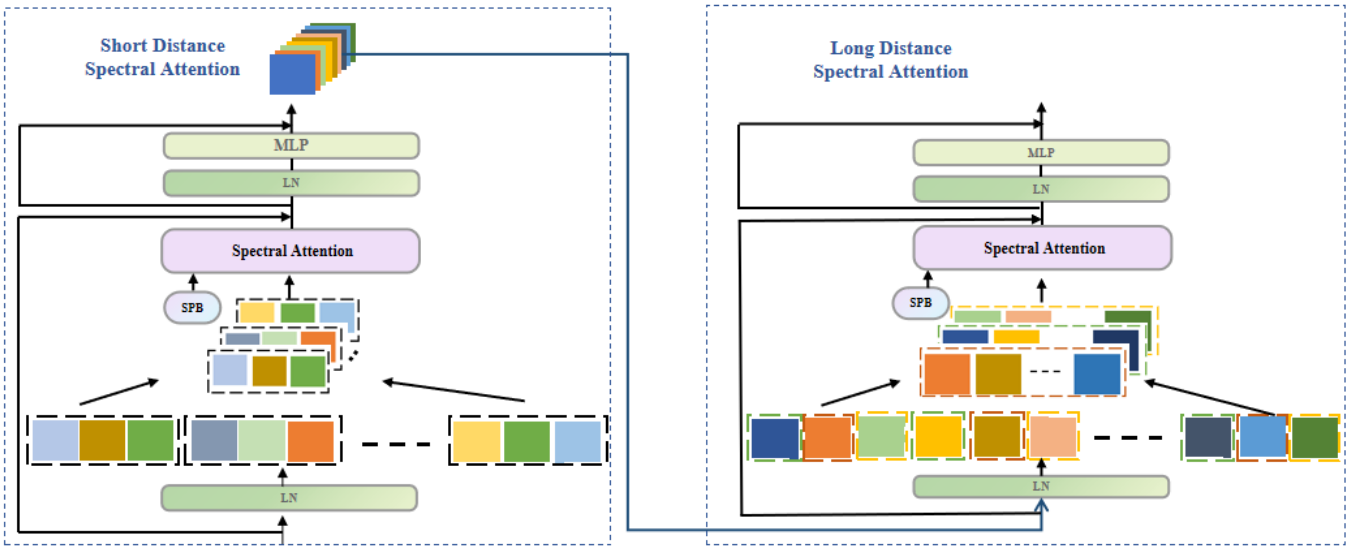


Fig. 4. Structure of the LSDSC module.

The long-distance grouping can be formulated as follows:

$$B'_L = \left(B'_1, B'_{1+T'}, \ldots, B'_{N-T'}\right) \ldots \left(B'_2, B'_{2+T'}, \ldots, B'_q\right), \left(B'_{\mathrm{T}}, B'_{T+T'}, \ldots, B'_N\right) \quad (5)$$

where B'_L denotes the long-distance spectral group and $(B'_2, B'_{2+T'}, \ldots, B'_q, \ldots)$ represents one of the long-distance spectral groups. B'_q denotes the feature of the qth spectral band after short-distance spectral embedding and T' denotes the sampling period.

3) Transformer Encoder: The Transformer encoder used in our proposed method comprises one spectral position bias (SPB), one multiheaded self-attention (MHSA), two LayerNorm (LN) layers, and one multilayer perceptron (MLP) layer.

Considering the varying number of spectral bands in different HSIs and the differing number of spectral groupings, the absolute position coding [24] is unsuitable for our proposed method. Therefore, we propose the SPB to generate the relative position between spectral bands when the spectral parameters of the test HSI do not match the trained model. The structure of SPB is shown in Fig. 5(a). The input ΔT_{ij} denotes the coordinate distance between the ith spectral band and the jth spectral band, and the output PB_{ij} denotes their encoded relative position feature. This nonlinear transformation consists of a normalization layer and an rectified linear unit (ReLU) activation, and the dimension of the intermediate layer is set to $D/4$, where D denotes the dimension of the embedding.

The spectral attention in Fig. 4 is achieved by the MHSA, which is illustrated in Fig. 5(b). The spectral attention can be calculated by

$$\text{Attention} = \text{Softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}} + \text{PB}\right)V \quad (6)$$

where Q, K, and $V \in \mathbb{R}^{G^2 \times D}$ denote the query, key, and value in the self-attentive module, respectively, $(d)^{1/2}$ is a constant regularizer, and PB represents the relative positions of the spectral bands.

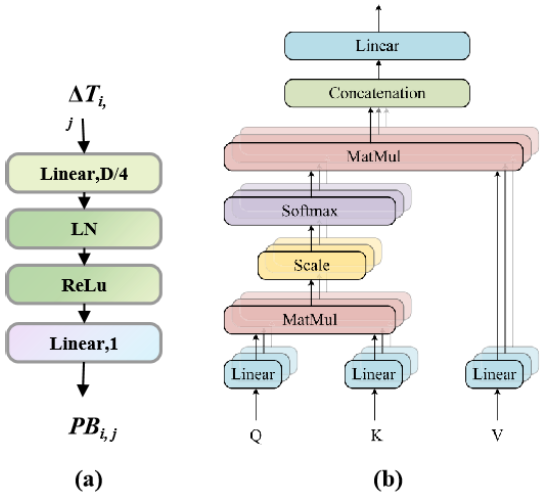


Fig. 5. (a) SPB. (b) MHSA.

For the MHSA, the original input features undergo a multigroup self-attention process, and then, each group of self-attention outputs is concatenated together with different semantic information. Subsequently, a linear transformation is performed to obtain the final output.

4) Dynamic Equilibrium Loss (DEL) Function: Since the HIC task belongs to multicategory classification and there is a category imbalance problem in the datasets, most existing HIC methods utilize the cross-entropy (CE) loss function [48] and the focal loss (FL) function [49]. However, there are two problems. On the one hand, the CE loss ignores the effect of category imbalance in the datasets. On the other hand, the category weight factor and the adjustment factor of the FL function are static, which is disadvantageous for parameter learning in the actual training process. Therefore, we propose a DEL function to alleviate these problems.

To dynamically adapt the model to the learning process at different stages of training, we propose adjusting the

TABLE I
INFORMATION ABOUT THREE HSI DATASETS

Dataset Name	Year	Equipment	Bands Number	Wavelength Range	Pixel Size	Categories Number
Indian Pines	1992	AVIRIS sensor	200	400-2500nm	145×145	16
Pavia University	2002	ROSIS sensor	103	0.43-0.86um	610×340	9
Salinas Valley	2003	AVIRIS sensor	204	400-2500nm	512×217	16

parameters according to the batch stage and the epoch stage. First, in the batch stage, the FL function of different categories should not be biased toward the majority category and ignore the minority category when the training is finally completed due to data imbalance. Therefore, we propose a strategy that makes the final loss of different categories in each batch the same, and as the model is trained, the model will gradually be biased toward the minority category. Second, in the epoch stage, to dynamically adjust the difficult and easy classification samples, we propose a strategy that makes the difficult and easy sample adjustment factors decrease with the increase of epochs, and as the model is trained, it does not lose its learning ability due to the decrease of the loss value.

The FL function is formulated as follows:

$$\mathrm{FL}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{7}$$

where t is the category of the current sample, α_t denotes the weight corresponding to the category t, and p_t denotes the probability value of the output probability distribution concerning the category t. The parameter γ is dynamically adjusted for different epochs

$$\mathrm{FL}(p_t)' = -\alpha_t(1-p_t)^{\frac{\alpha}{\sqrt{\beta}}}\log(p_t) \tag{8}$$

where β denotes the number of epochs for the current training.

The final loss function for category c_k within each batch is calculated by

$$\begin{aligned} F_{c_k} &= \sum_{i=1}^{n_{c_k}} \mathrm{FL}\left(p_i^{c_k}\right)' \\ &= \sum_{i=1}^{n_{c_k}} -\alpha_{c_k}\left(1-p_i^{c_k}\right)^{\frac{\gamma}{\sqrt{\beta}}}\log\left(p_i^{c_k}\right) \end{aligned} \tag{9}$$

where m denotes the number of categories covered by the current datasets, i denotes the current sample, c_k denotes the category of the current sample, and n_{ck} denotes the number of samples of category c_k in the current batch.

According to the principle that the final loss is the same across different categories within each batch, the formula is as follows:

$$F_{(c_1)} = F_{(c_2)} = \cdots = F_{(c_k)}. \tag{10}$$

In the ideal case where the above conditions hold, although the actual training process cannot reach that ideal state, with the training of the model, the model will eventually be biased toward focusing on a few classes. So that the final training results are independent of the number of samples, thus reducing the impact of imbalance in the categories of the dataset.

E. Implementation

To better illustrate the proposed LSDnet, we provide a detailed description using the Pavia University (PU) dataset as an example. The size of the dataset is $610 \times 340 \times 103$. After patch extraction, the size of each patch is $11 \times 11 \times 103$. In the first 3-D convolutional layer, 8 feature cubes of size $9 \times 9 \times 101$ are generated through convolutional operations, using 8 cubic kernels of size $3 \times 3 \times 3$ for each block. The 3-D convolution is used in this step because each patch contains rich spectral information. The eight feature cubes are then rearranged to form a single feature cube of size $9 \times 9 \times 808$. Following that, 2-D convolution with 49 3×3 plane kernels produces 49 feature maps, each of size 7×7. This output, $A_{\text{in}} \in R^{7\times7\times49}$, simultaneously serves as the input for the BFFE module. Also, the BFFE module does not change the size of the features.

In the next step, to facilitate data processing, the previously obtained $A_{\text{out}} \in R^{7\times7\times49}$ undergoes linear mapping, yielding a feature $X \in R^{48\times48}$. We then pass these features through the MSSE module to obtain a feature T of size $12 \times 12 \times 96$. Each feature map is flattened into a 1-D feature vector, resulting in a vector of size 1×144. At this point, the features obtained are equivalent to $L_{\text{in}} \in R^{96\times144}$. Following the LSDSC module, the feature size remains unchanged.

III. EXPERIMENTAL RESULTS AND ANALYSIS

We have verified the effectiveness and advancement of our proposed method on three typical publicly available datasets, including the Indian Pines (IP), PU, and Salinas Valley (SV) datasets. The details of these datasets are listed in Table I. Furthermore, we have validated the contributions of the BFFE module, DEL function, SPB module, and LSDSC module through ablation experiments.

A. Experimental Setup

1) *Evaluation Metrics:* Four evaluation metrics were employed to quantitatively evaluate the classification performance: total accuracy (OA), average accuracy (AA), kappa coefficient (K), and the classification accuracy of each land cover category. For each metric, higher values indicate better classification performance.
2) *Configuration:* The following computer setups were used to carry out the validation tests for the proposed approach: an i7-8700 K CPU, 32-GB RAM, and a GTX1080 GPU. We set the starting learning rate, epoch count, and batch size to $4e-3$, 100, and 64, respectively.
3) *Comparing methods:* To verify the advancement of our proposed method, we employ eight HIC methods as comparisons, including 2DCNN [50], 3DCNN [16],

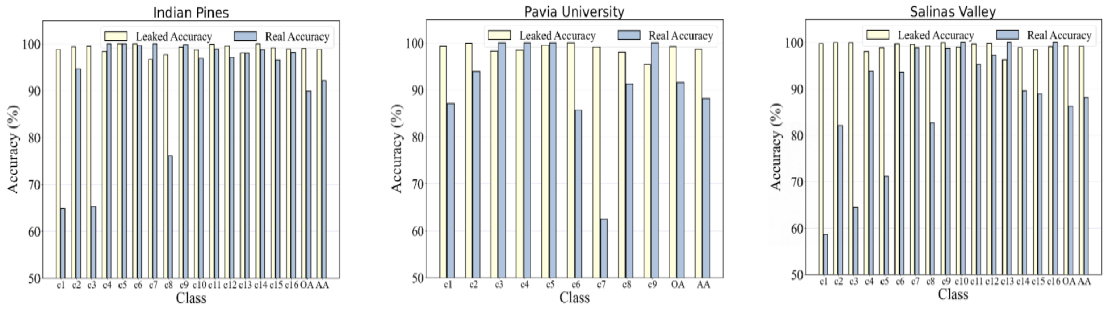


Fig. 6. Influence of the information leakage on nonleaked and leaked datasets.

depthwise separable convolution neural network (DSP) [51], deep feature fusion network (DFFN) [52], bidirectional long short-term memory network (BLSTM) [21], SpectralFormer (SF) [35], group-aware hierarchical transformer (GF) [34], spectral–spatial feature tokenization Transformer (SSFTT) [33], and dual-branch multiscale Transformer (DBMST) [40]. The network configurations were identical to those in the corresponding references.

B. Robust Data Partitioning

The data partitioning strategy directly affects the robustness and reasonableness of the experimental results. However, most existing HIC methods may suffer from information leakage between test data and training data in their experiments [41], because they sample the training and testing data by central pixel, which may lead to overlap between the training patch and the test patch. Meanwhile, the imbalance problem between different categories will also affect the classification accuracy for each category of target.

To verify the necessity of rationally partitioning the datasets, we tested the SSFTT [33] on three datasets using two data partitioning strategies, respectively, and the test classification results are displayed in Fig. 6. The leaked accuracy indicates the classification result on the leaked datasets, which are constructed by selecting HSI patches pixel-by-pixel. The real accuracy indicates the classification result on the datasets without information leakage, which are constructed by our proposed data partitioning strategy. After dividing the datasets without information leakage, the OA and AA metrics of SSFTT on the three databases were significantly decreased. Meanwhile, specific to the classification accuracy of each category, we found that the unbalanced data partitioning method caused a significant drop in classification accuracy for some categories.

To solve the problems of information leakage and category imbalance, we propose a category-by-category data partitioning method to obtain HSI patches without information leakage. Taking category j as an example, we select the HSI patches for category j following these rules. First, we randomly select a pixel of category j as the central pixel of an image patch, and this patch is the initial patch of category j. Second, we select another pixel of category j and judge whether the image patch determined by it overlaps with the initial patch. If there is an overlap, this pixel is discarded. Otherwise, this new image patch is added to the data of category j. Then, we follow this rule to select more patches and ensure that they do not overlap with all established patches of category j until the number of patches of category j reaches a predetermined number S_j. Finally, in order to reduce the impact of category imbalance, we propose to utilize an adjustment factor to adjust the number of patches of each category by decreasing the proportion of majority categories and increasing the proportion of minority categories. The number of patches of each class can be expressed as follows:

$$S_j = \alpha \times \lfloor \frac{(H \times W)}{t^2} \rfloor \times \frac{m_j + p_j}{m_1 + m_2 + \cdots + m_k} \tag{11}$$

$$p_j = \begin{cases} p_{\max}, & if \quad m_j \geq \frac{1}{k} \times \left(\sum_{j=1}^{k} m_j \right) \\ p_{\min}, & if \quad m_j \leq \frac{1}{k} \times \left(\sum_{j=1}^{k} m_j \right) \end{cases} \tag{12}$$

where S_j represents the number of patches of category j, with $j = 1, 2, 3, \ldots, k$, and k being the number of categories in the HSI dataset. W and H denote the width and height of the HSI, respectively. Moreover, m_j is the pixel number of category j in the original HSI and t is the spatial size of an HSI patch. $\lfloor ((H \times W)/t^2) \rfloor$ denotes the maximum number of nonoverlapping HSI patches that can be selected. In addition, α denotes the adjustment factor for the number of nonoverlapping HSI patches. p_j denotes the scaling factor for each category, with a small scaling factor $p_{\min}$ for the majority categories and a large scaling factor $p_{\max}$ for the minority categories.

When selecting HSI patches without information leakage, some image patches with information leakage are discarded, resulting in a decrease in the actual amount of usable data. Therefore, to ensure sufficient training of the model, the training set and testing set are divided for each category in a ratio of 3:7. Furthermore, we employ augmentation methods [53] on both the training set and testing set to improve the diversity of the data.

C. Parameter Analysis

First, the size of the HSI patch is a key parameter in HIC methods, because a small size is not conducive to extracting sufficient spatial information, and a large size may lead to category inconsistency within the image patch. Based on our experience, we set the patch size to $5, 7, 9, 11$, and 13 for

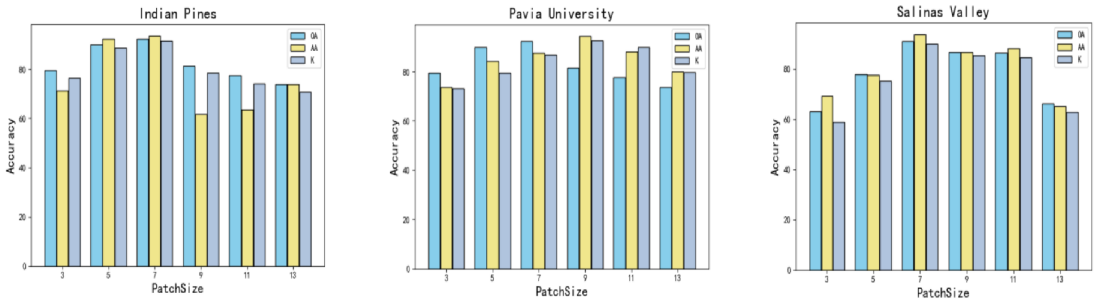


Fig. 7. OA, AA, and K metrics of different patch sizes on three datasets.

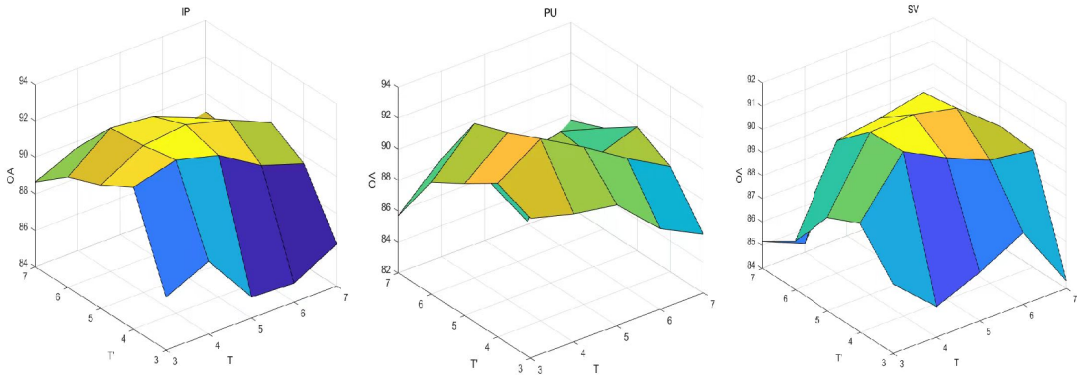


Fig. 8. Influence of the length of short-distance spectral group and the interval sampling length of the long-distance spectral group.

each of the three databases, and the classification results are shown in Fig. 7. It can be observed that when the patch size is set to 7, high classification accuracy is achieved on all three databases. Therefore, the patch size is set to 7 in our proposed method.

Second, the length of the short-distance spectral group and the interval sampling length of the long-distance spectral group are two important parameters in our proposed LSDSC module. We set the lengths of the short-distance spectral group to 3–7, and the interval sampling lengths of the long-distance spectral group to 3–7. A grid search was performed on three datasets to find the optimal parameters, and the OA metrics are displayed in Fig. 8. By comparing the experiments on the three datasets, both of these parameters are set to 4.

Third, we propose a dynamically adjustable parameter γ in the DEL function. We set it to $10, 20, 30$, and 40, and 50 separately, and the corresponding OA metrics on the three datasets are demonstrated in Fig. 9. Based on the experimental results, we set γ to 20 in our comparative experiments.

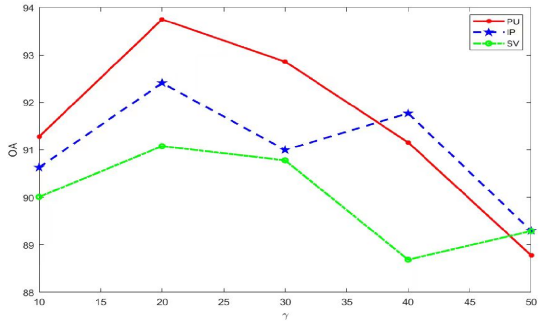


Fig. 9. Influence of the parameter γ in the loss function on three datasets.

D. Impact of Training Samples

To verify the impact of training data quantity on our proposed method, we compared all comparison methods using five different proportions of training data. We selected $1\%, 5\%$, and 10%, and $15\%, 20\%$, and 30% of the overall samples from the IP, Salinas, and PU datasets for training. The OA scores of the three methods on the three datasets are shown in Fig. 10. As the training data increases, the overall accuracy of the three methods shows an upward trend. Meanwhile, at each training ratio, our method consistently achieved the optimal classification accuracy. Therefore, taking into account the classification accuracy and the difficulty of obtaining training data for HSI, we have selected 30% of the data as our training data.

E. Ablation Study

To evaluate the performance of each module in the proposed methods, we carried out the following ablation tests.

1) BFFE Module: To test the effectiveness of the BFFE module, we designed three comparison methods, including feature enhancement using the GF, feature enhancement using the BF, and no feature enhancement (NO) on three datasets. The experimental results are shown in Table II. We can find that the BF-based method achieved the highest OA, AA, and K metrics on the three datasets.

2) MSSE Module: To verify the performance of the MSSE module, we tested the strategy of no multiscale module (NO)

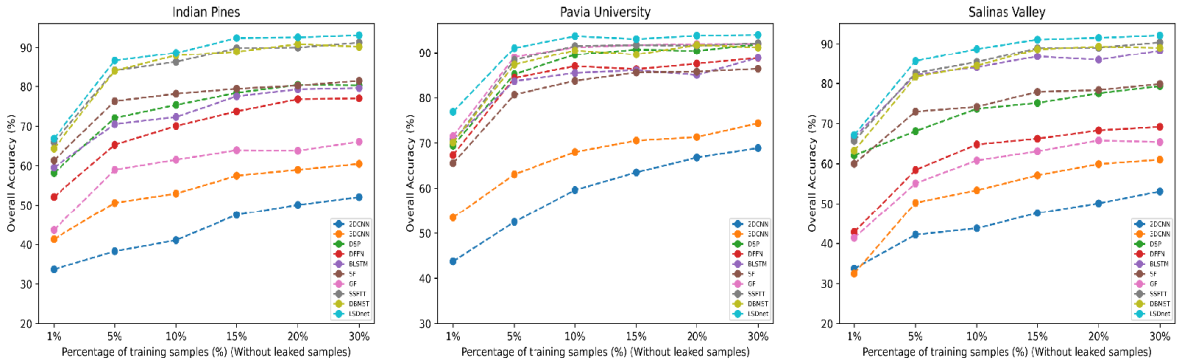


Fig. 10. OA scores (%) of all comparison methods on three datasets with 1%, 5%, 10%, 15%, 20%, and 30% training samples per category.

TABLE II
ABLATION ANALYSIS FOR THE BFFE MODULE ON THREE DATASETS (THE BEST RESULTS ARE BOLDED)

Metric	PU			IP			SV		
	BF	GF	NO	BF	GF	NO	BF	GF	NO
OA	**93.75**	92.25	92.01	**92.41**	91.55	91.27	**91.08**	89.33	89.21
AA	**94.28**	91.70	91.54	**93.30**	92.16	90.88	**93.62**	92.08	90.51
K	**92.49**	91.55	90.78	**91.50**	89.93	90.31	**90.05**	90.01	89.10

TABLE III
ABLATION ANALYSIS FOR THE MSSE MODULE ON THREE DATASETS (THE BEST RESULTS ARE BOLDED)

Metric	PU		IP		SV	
	MSSE	NO	MSSE	NO	MSSE	NO
OA	**93.75**	91.56	**92.41**	90.00	**91.08**	90.83
AA	**94.28**	89.35	**93.30**	91.78	**93.62**	89.91
K	**92.49**	91.66	**91.50**	90.26	**90.05**	89.44

TABLE IV
ABLATION ANALYSIS FOR THE LSDSC MODULE ON THREE DATASETS (THE BEST RESULTS ARE BOLDED)

Metric	PU		IP		SV	
	LSDSC	ViT	LSDSC	ViT	LSDSC	ViT
OA	**93.75**	92.03	**92.41**	90.45	**91.08**	87.75
AA	**94.28**	92.88	**93.30**	91.23	**93.62**	91.85
K	**92.49**	90.73	**91.50**	90.08	**90.05**	89.69

on the three datasets, respectively. The results in Table III show that our proposed MSSE module effectively incorporated important information between different levels and achieved the best performance on the three datasets.

3) LSDSC Module: To verify the ability of the LSDSC module, we compared the performance of the proposed method on three datasets when using the LSDSC module and classical ViT, respectively. The experimental results shown in Table IV demonstrate the effectiveness of our proposed LSDSC module.

4) DEL Function: To evaluate the performance of the DEL function, we employ three loss functions for comparison, including the proposed DEL, the FL, and the CE loss. The results shown in Table V demonstrate that our proposed DEL-based method achieved the best OA, AA, and K metrics on all three datasets.

F. Classification Results and Analysis

1) Quantitative Classification Results: The complete classification results are displayed in Tables VI–VIII. The comparison results mainly refer to the metrics OA, AA, and K. In general, the CNN architecture demonstrated its effectiveness in extracting spectral features, and the Transformer architecture demonstrated a powerful sequence feature extraction capability.

Specifically, the networks based on 2DCNN and 3DCNN were unable to fully extract the features of HSI and exhibited only low classification accuracy. We analyzed the possible reasons as follows. On the one hand, the strong complementary but related information between different layers of HSI was ignored, and DFN achieved fusion between different layers by introducing residual learning, exhibiting higher accuracy than 3DCNN. On the other hand, the HIC model could not be deepened. In this regard, the DSP network combined residual connectivity and deep separable convolution to construct residual units with a small number of parameters to deepen the network and further improve classification accuracy.

Although the Transformer has the power to extract feature sequences, it cannot obtain the local differences of spectral features. Therefore, the GF network was proposed to achieve better classification performance compared with the ViT-based SF network through the combination of CNN and Transformer. Tables VI–VIII show the performance of our proposed LSDnet on the three datasets. In comparison with the newer SSFTT method, our method improved OA, AA, and K by 2.2%, 6.13%, and 2.56% on the PU dataset, by 2.59%, 2.04%, and 2.89% on the IP dataset, and by 2.17%, 3.04%, and 2.46% on the SV dataset, respectively. Overall, the experimental results demonstrated the relative strength of the generalization ability and performance stability of our proposed LSDnet.

Due to the unbalanced sample of the dataset, the existing methods were not effective for the categories 'Bare Soil',

TABLE V
ABLATION ANALYSIS FOR THE DEL FUNCTION ON THREE DATASETS (THE BEST RESULTS ARE BOLDED)

Metric	Pavia University			Indian Pines			Salinas Valley		
	DBL	FL	CE	DEL	FL	CE	DEL	FL	CE
OA	**93.75**	93.00	93.15	**92.41**	92.16	91.55	**91.08**	90.66	90.53
AA	**94.28**	93.65	92.86	**93.30**	92.85	92.38	**93.62**	92.77	91.65
K	**92.49**	91.08	90.59	**91.50**	90.46	89.22	**90.05**	89.39	90.26

TABLE VI
EXPERIMENTAL RESULTS OF ALL COMPARING METHODS ON THE PAVIA DATASET (THE BEST RESULTS ARE BOLDED)

Class	2DCNN	3DCNN	DSP	DFFN	BLSTM	SF	GF	SSFTT	DBMST	LSDnet
C1	67.37	76.84	93.94	86.31	90.52	87.37	**95.79**	90.00	95.08	87.87
C2	47.88	66.59	97.30	94.65	**99.77**	99.55	90.65	96.87	87.49	97.30
C3	23.33	27.78	72.73	80.00	76.67	66.67	56.67	60.00	**95.84**	90.91
C4	26.67	36.84	90.48	93.33	86.67	86.67	86.67	**100.00**	99.25	**100.00**
C5	61.90	47.62	**100.00**	66.67	85.71	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
C6	98.48	81.81	55.56	77.27	46.97	41.67	96.21	75.00	**100.00**	77.78
C7	63.33	73.33	50.00	33.33	83.33	40.00	96.67	71.43	22.02	**100.00**
C8	87.50	79.17	94.74	87.50	87.50	66.67	91.67	**100.00**	93.79	94.74
C9	**100.00**	75.00	**100.00**	**100.00**	**100.00**	75.00	75.00	**100.00**	99.85	**100.00**
OA	59.62 ± 2.23	68.02 ± 2.37	89.58 ± 1.09	87.00 ± 1.83	87.50 ± 1.37	83.75 ± 2.68	91.25 ± 0.77	91.55 ± 1.23	90.56 ± 1.02	$\mathbf{93.75 \pm 1.09}$
AA	64.05 ± 3.56	62.76 ± 2.99	82.62 ± 2.30	79.87 ± 1.07	84.12 ± 1.75	73.73 ± 1.60	87.70 ± 1.83	88.15 ± 1.63	88.14 ± 1.72	$\mathbf{94.28 \pm 1.48}$
K	44.64 ± 0.91	55.24 ± 1.15	87.31 ± 0.80	79.84 ± 0.79	79.52 ± 0.38	73.06 ± 0.46	86.73 ± 0.37	89.93 ± 0.29	87.74 ± 0.77	$\mathbf{92.49 \pm 0.32}$

TABLE VII
EXPERIMENTAL RESULTS OF ALL COMPARING METHODS ON THE IP DATASET (THE BEST RESULTS ARE BOLDED)

Class	2DCNN	3DCNN	DSP	DFFN	BLSTM	SF	GF	SSFTT	DBMST	LSDnet
C1	27.27	33.33	66.67	90.90	0.00	54.55	27.27	75.00	59.37	**91.17**
C2	44.17	61.76	77.05	39.39	87.76	72.85	19.69	81.07	**95.30**	84.89
C3	27.76	68.45	74.76	90.22	35.96	79.18	**96.52**	78.55	75.56	84.86
C4	13.92	24.05	74.68	78.48	46.84	68.35	75.94	87.34	98.19	**100.00**
C5	31.65	41.17	45.38	26.58	83.54	89.87	84.81	92.86	**97.34**	97.06
C6	67.29	62.41	84.96	75.94	92.48	96.61	83.46	96.24	98.04	**100.00**
C7	6.25	31.25	96.55	81.25	62.50	87.50	81.25	**100.00**	90.00	97.41
C8	45.30	59.06	82.46	84.53	91.16	99.45	99.45	94.15	**100.00**	87.72
C9	0.00	16.67	33.33	33.33	0.00	16.67	33.33	**100.00**	57.14	83.33
C10	37.74	73.05	59.29	83.29	92.18	86.25	53.90	94.34	93.82	**96.77**
C11	62.34	47.26	89.45	73.73	86.81	77.85	35.02	88.09	75.91	**93.99**
C12	11.06	55.75	91.15	74.78	53.10	51.77	92.92	**95.58**	90.36	91.15
C13	56.79	50.62	87.65	70.37	54.32	67.90	87.65	97.50	99.30	**100.00**
C14	68.94	77.37	77.37	**96.48**	76.19	93.99	91.48	92.84	92.55	90.91
C15	21.14	18.69	67.48	89.43	58.54	39.84	91.87	91.87	**100.00**	93.49
C16	34.21	47.37	**100.00**	94.74	94.74	57.89	84.21	94.74	**100.00**	**100.00**
OA	47.53 ± 2.56	57.50 ± 3.01	78.42 ± 1.89	73.74 ± 2.3	77.55 ± 1.78	79.46 ± 1.99	63.94 ± 2.15	89.82 ± 1.45	88.91 ± 2.31	$\mathbf{92.41 \pm 1.39}$
AA	34.74 ± 1.25	48.02 ± 1.68	75.51 ± 0.99	73.96 ± 0.81	63.50 ± 0.65	71.28 ± 0.85	71.48 ± 1.13	91.26 ± 0.86	88.93 ± 1.52	$\mathbf{93.30 \pm 0.68}$
K	39.33 ± 2.79	52.65 ± 3.31	75.69 ± 2.20	70.67 ± 2.15	74.23 ± 1.83	76.43 ± 1.92	60.46 ± 1.68	88.61 ± 1.99	87.46 ± 1.90	$\mathbf{91.50 \pm 1.76}$

'Bitumen', 'Alfalfa', "Buildings," "Grass," "Trees," "Lettuce 6wk," and "Lettuce 7wk," which exhibited lower classification precision. Our LSDnet can improve the classification accuracy of these categories with fewer training samples to a certain extent by focusing on important positions in HSI to obtain better classification performance. For example, on the SV dataset, LSDnet improved the category "Lettuce 4wk" by 16.92% compared with SSFTT.

2) Visualization Evaluation: In Figs. 11–13, we depict the classification results from the tests performed on the three datasets. The statistics show that, in comparison with other methods, the classification maps produced by our LSDnet are the cleanest and most accurate. In the IP dataset, the yellow area is harder to distinguish and is easily misclassified as brown. However, the proposed method can identify the yellow area to a greater extent, thus proving the excellent performance of the LSDnet.

The Transformer can obtain dependencies between long-range features and focus more on low-frequency features. Therefore, it is unable to classify small-size features well. The ViT-based SF network misclassifies the categories "Bitumen," "Alfalfa," and "Weeds 1." To this end, we suggest a network, GF with SSFTT, based on CNN with Transformer, and it can be observed that the obtained classification achieves better results in the categories "Bitumen" and "Alfalfa." For our proposed LSDnet with a DEL function, it can be shown that the classification performance is comparatively stable for some categories with less data volume.

3) Model Complexity Analysis: Table IX lists the number of parameters, floating-point operations per seconds (FLOPs), training time, and testing time for the comparison methods on these three datasets. It can be observed that most CNN-based methods have fewer parameters and FLOPs, along with

TABLE VIII
EXPERIMENTAL RESULTS OF ALL COMPARING METHODS ON THE SV DATASET (THE BEST RESULTS ARE BOLDED)

Class	2DCNN	3DCNN	DSP	DFFN	BLSTM	SF	GF	SSFTT	DBMST	LSDnet
C1	65.34	98.29	95.45	72.16	96.02	**100.00**	92.05	98.59	69.06	99.43
C2	67.74	90.02	**100.00**	97.65	87.39	89.44	56.01	99.65	88.49	82.99
C3	17.22	**100.00**	95.56	6.11	96.67	67.22	90.00	63.21	99.28	99.44
C4	31.96	95.90	91.80	81.15	93.44	94.26	94.26	96.28	89.24	**99.18**
C5	30.42	72.92	**100.00**	72.50	89.58	95.83	65.00	97.71	89.33	81.67
C6	57.78	96.11	85.83	96.94	98.33	98.61	42.50	**100.00**	81.95	99.72
C7	50.46	82.15	67.08	61.23	69.54	**100.00**	69.85	98.75	99.80	97.23
C8	62.35	80.93	90.17	88.62	86.67	98.35	85.21	93.24	**99.77**	88.23
C9	60.45	5.53	57.17	0.35	**99.83**	85.84	60.10	99.26	94.67	96.89
C10	33.57	36.13	65.70	86.50	84.31	37.23	61.31	**96.37**	95.86	94.89
C11	14.44	61.11	44.44	25.56	70.00	20.00	51.11	74.19	**93.58**	91.11
C12	8.19	13.45	67.25	21.05	97.66	99.42	87.72	99.08	97.10	**100.00**
C13	5.71	27.14	4.29	81.43	64.29	**100.00**	88.57	98.59	**100.00**	**100.00**
C14	27.16	29.40	22.22	87.65	**98.77**	46.91	65.43	97.69	93.51	91.36
C15	38.98	3.14	55.92	65.22	72.71	16.79	8.09	51.03	59.37	**77.81**
C16	37.42	67.48	67.48	**98.16**	82.21	92.64	91.41	85.61	84.18	**98.16**
OA	47.62 ± 2.50	57.09 ± 3.11	75.14 ± 2.21	66.24 ± 1.88	86.78 ± 1.53	77.99 ± 1.09	63.11 ± 1.39	88.91 ± 1.46	88.65 ± 1.58	$\mathbf{91.08 \pm 1.25}$
AA	38.07 ± 1.71	59.92 ± 2.01	69.39 ± 1.95	65.14 ± 0.73	86.71 ± 1.38	77.65 ± 1.22	69.28 ± 1.01	90.58 ± 0.93	89.70 ± 1.10	$\mathbf{93.62 \pm 0.89}$
K	41.07 ± 0.94	52.70 ± 1.57	72.43 ± 1.39	62.73 ± 0.96	85.27 ± 0.78	75.20 ± 0.55	59.02 ± 0.36	87.59 ± 0.51	87.32 ± 0.62	$\mathbf{90.05 \pm 0.27}$

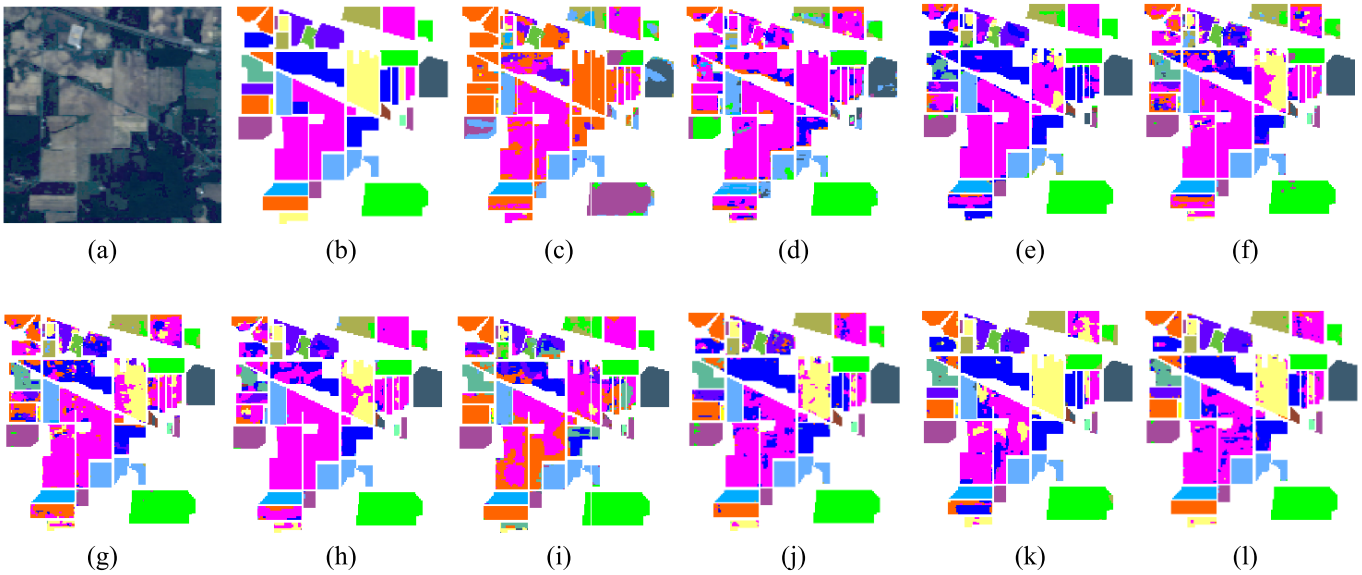


Fig. 11. Classification maps of all comparison methods and our proposed LSDnet on IP datasets. (a) False color image. (b) Ground truth. (c) 2-D. (d) 3-D. (e) DSP. (f) DFFN. (g) BLSTM. (h) SF. (i) GF. (j) SSFTT. (k) DBMST. (l) LSDnet.

TABLE IX
NUMBER OF PARAMETERS (K), FLOPS (10E6), TRAINING TIME, AND TESTING TIME OF DIFFERENT ALGORITHMS

Moules	PU				IP				SV			
	Parameters	Flops	Train(s)	Test(s)	Parameters	Flops	Train(s)	Test(s)	Parameters	Flops	Train(s)	Test(s)
2DCNN	**81**	**234**	**31.2**	**1.72**	156	453	**13.8**	**0.60**	154	462	**51.6**	3.10
3DCNN	141	2272	62.3	2.52	395	4415	27.6	0.95	395	4501	120.9	5.68
DSP	250	454	58.4	3.65	254	473	32.3	1.23	254	474	185.2	11.05
DFFN	240	861	181.6	4.70	217	741	50.8	1.48	217	742	241.0	7.36
BLSTM	518	1250	158.6	5.04	1578	3076	59.8	1.90	1635	3165	295.3	10.70
SF	170	964	327.0	15.41	348	2345	98.8	5.56	358	2411	491.2	27.39
GF	927	4804	162.7	5.65	972	5036	43.6	1.34	972	5036	213.0	8.92
SSFTT	149	**234**	69.0	2.14	149	**234**	21.0	**0.60**	198	**234**	82.8	**2.79**
DBMST	137	1043	732.83	3.47	**137**	1043	169.47	0.96	**137**	1043	828.35	5.26
LSDnet	584	3324	165.0	6.34	623	4305	46.8	1.76	623	4305	220.8	10.00

shorter training and testing times. However, their performance is inferior. Compared with those Transformer-based methods, the proposed LSDnet has more parameters and higher FLOPs, only slightly less than GF. Nevertheless, it outperforms Transformer-based algorithms significantly within reasonable testing times. Therefore, this is an aspect that needs to be

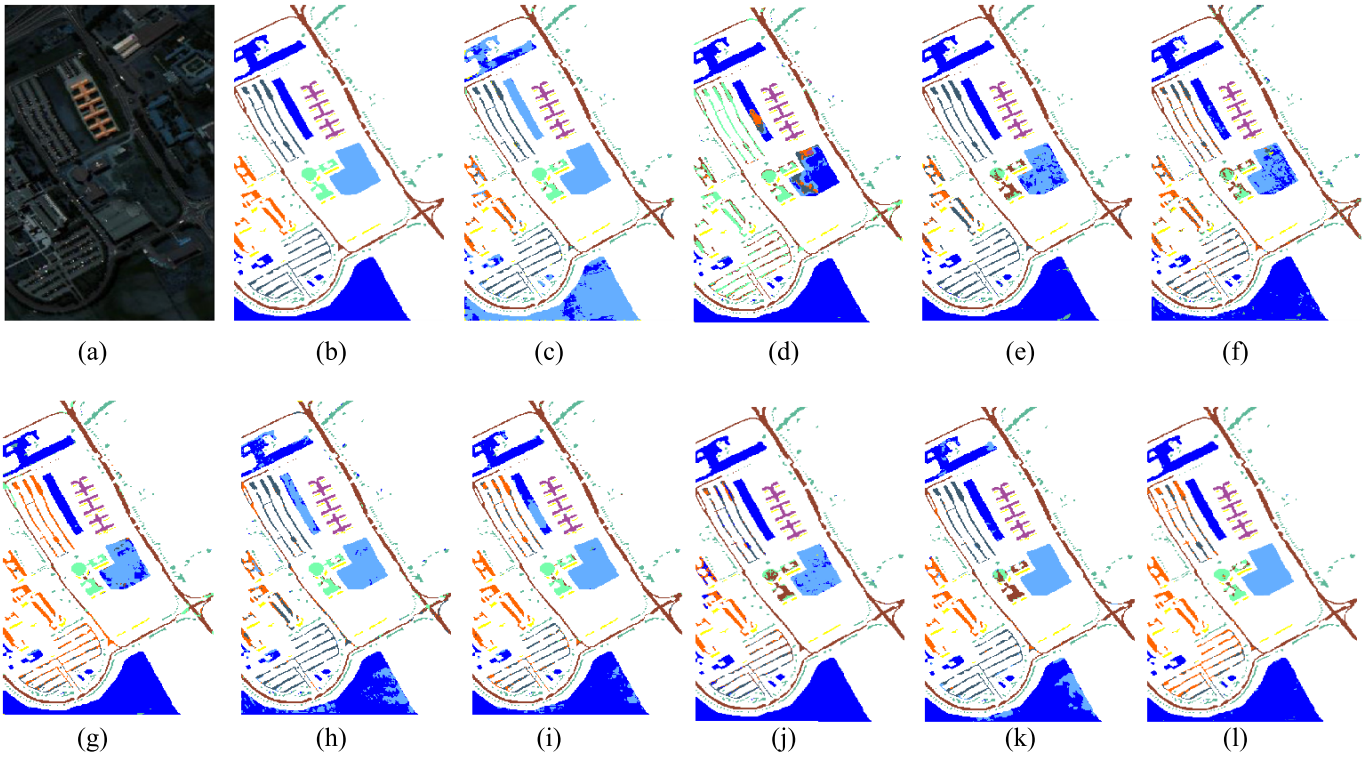


Fig. 12. Classification maps of all comparison methods and our proposed LSDnet on PU datasets. (a) False color image. (b) Ground truth. (c) 2-D. (d) 3-D. (e) DSP. (f) DFFN. (g) BLSTM. (h) SF. (i) GF. (j) SSFTT. (k) DBMST. (l) LSDnet.

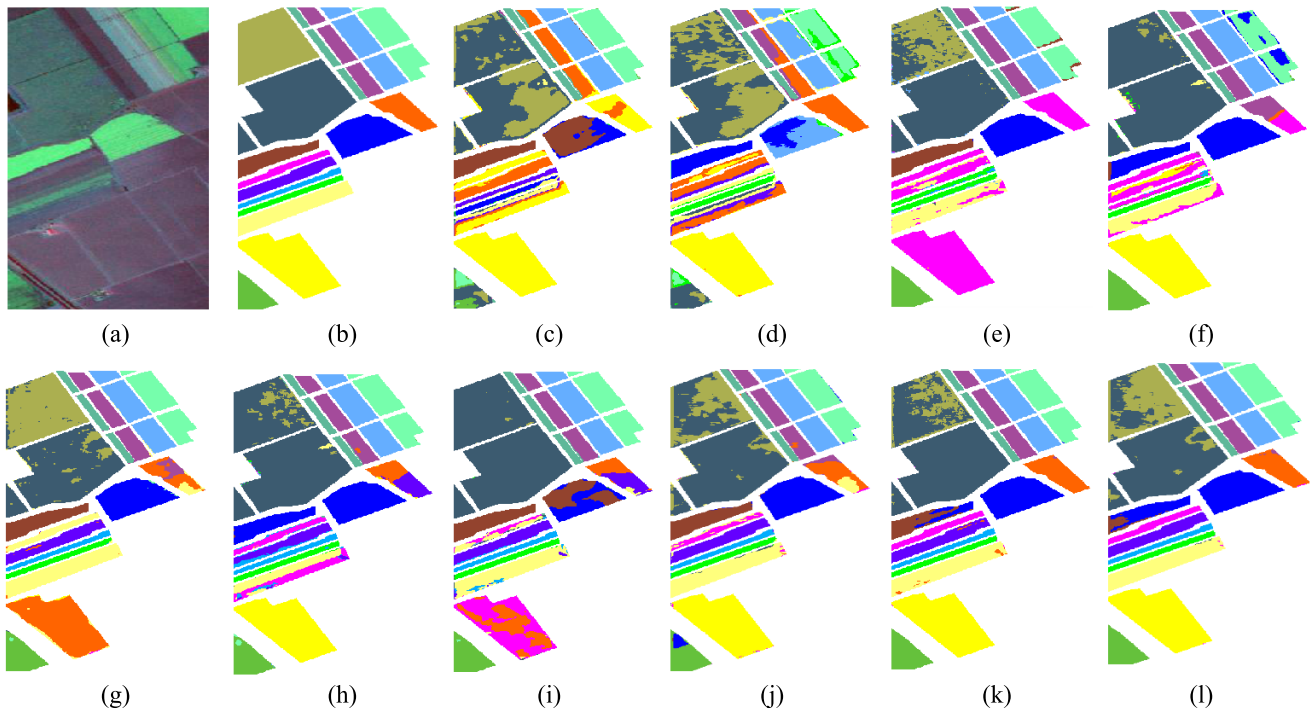


Fig. 13. Classification maps of all comparison methods and our proposed LSDnet on SV datasets. (a) False color image. (b) Ground truth. (c) 2-D. (d) 3-D. (e) DSP. (f) DFFN. (g) BLSTM. (h) SF. (i) GF. (j) SSFTT. (k) DBMST. (l) LSDnet.

improved in our future research. Considering the complexity of the LSDnet, we need to further investigate lightweight models in the future to promote the practical application of HIC and accelerate inference times.

IV. Conclusion

Transformer-based models have very powerful performance in HSI image classification. In this article, we have proposed

the LSDnet model based on Transformer. With the help of MSSE, long- and short-distance spectral cross attention can effectively explore both coarse-grained and fine-grained features. Meanwhile, we have designed a dynamic equalization loss function and a BFFE to reduce the impact of category imbalance and interfering pixels within the HSI patch, respectively. Furthermore, we have discussed the impact of the data partitioning strategy on the robustness of HSI methods and proposed a new data partitioning strategy to prevent the information leakage problem. The experimental results demonstrated that our proposed network outperforms other well-known and conventional HIC methods. The ablation study also showed that the proposed individual modules all enhanced the classification accuracy to different extents.

In our future work, we will design more efficient cross-scale spectral embedding to fuse spectral information from different scales. In addition, we will study HSI generation based on a GAN network for dataset expansion and then improve the hyperspectral classification accuracy.

REFERENCES

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] Y. Qian, M. Ye, and J. Zhou, "Hyperspectral image classification based on structured sparse logistic regression and three-dimensional wavelet texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 4, pp. 2276–2291, Apr. 2013.

[3] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[4] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[5] N. Yokoya, J. Chan, and K. Segl, "Potential of resolution-enhanced hyperspectral data for mineral mapping using simulated EnMAP and Sentinel-2 images," *Remote Sens.*, vol. 8, no. 3, p. 172, Feb. 2016.

[6] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdiscipl. Rev., Comput. Statist.*, vol. 2, no. 4, pp. 433–459, Jul. 2010.

[7] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[8] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[9] S. K. Roy, S. Das, T. Song, and B. Chanda, "DARecNet-BS: Unsupervised dual-attention reconstruction network for hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 12, pp. 2152–2156, Dec. 2021.

[10] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[11] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[12] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[13] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.

[14] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[15] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Jan. 2015.

[16] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[17] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

[18] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[19] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[20] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2017.

[21] S. Mei, X. Li, X. Liu, H. Cai, and Q. Du, "Hyperspectral image classification using attention-based bidirectional long short-term memory network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 1, pp. 1–21, Jun. 2021.

[22] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.

[23] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.

[24] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[25] B. Heo, S. Yun, D. Han, S. Chun, J. Choe, and S. J. Oh, "Rethinking spatial dimensions of vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 11936–11945.

[26] B. Graham et al., "LeViT: A vision transformer in ConvNet's clothing for faster inference," 2021, *arXiv:2104.01136*.

[27] D. Zhou et al., "DeepViT: Towards deeper vision transformer," 2021, *arXiv:2103.11886*.

[28] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2021, pp. 558–567.

[29] X. Chen, S. Kamata, and W. Zhou, "Hyperspectral image classification based on multi-stage vision transformer with stacked samples," in *Proc. IEEE Region 10 Conf. (TENCON)*, Dec. 2021, pp. 441–446, doi: 10.1109/TENCON54134.2021.9707289.

[30] X. He, Y. Chen, and Q. Li, "Two-branch pure transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415, doi: 10.1109/TGRS.2022.3203476.

[32] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[33] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[34] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[35] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, no. 4, pp. 1–23, May 2021.

[36] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[37] J. Wang, M. Zhang, W. Li, and R. Tao, "A multistage information complementary fusion network based on flexible-mixup for HSI-X image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 1, no. 2, pp. 1–13, Aug. 2024.

[38] Y. Gao, W. Li, J. Wang, M. Zhang, and R. Tao, "Relationship learning from multisource images via spatial-spectral perception network," *IEEE Trans. Image Process.*, vol. 33, pp. 3271–3284, 2024.

[39] Z. Meng, Q. Yan, F. Zhao, and M. Liang, "Multi-scale feature attention and transformer for hyperspectral image classification," in *Proc. 13th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Oct. 2023, pp. 1–5.

[40] C. Shi, S. Yue, and L. Wang, "A dual-branch multiscale transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504520.

[41] L. Zou, X. Zhu, C. Wu, Y. Liu, and L. Qu, "Spectral–spatial exploration for hyperspectral image classification via the fusion of fully convolutional networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, no. 3, pp. 659–674, Apr. 2020.

[42] L. Qu, X. Zhu, J. Zheng, and L. Zou, "Triple-attention-based parallel network for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 2, p. 324, Jan. 2021.

[43] D. Liao, C. Shi, and L. Wang, "A complementary integrated transformer network for hyperspectral image classification," *CAAI Trans. Intell. Technol.*, vol. 8, no. 4, pp. 1288–1307, Dec. 2023.

[44] P. Singhal, A. Verma, and A. Garg, "A study in finding effectiveness of Gaussian blur filter over bilateral filter in natural scenes for graph based image segmentation," in *Proc. 4th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, Jan. 2017, pp. 1–6.

[45] H. Fan et al., "Multiscale vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 6824–6835.

[46] C.-F. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, May 2021, pp. 357–366.

[47] X. Guo, X. Zhang, L. Li, and Z. Xia, "Micro-expression spotting with multi-scale local transformer in long videos," *Pattern Recognit. Lett.*, vol. 168, pp. 146–152, Apr. 2023.

[48] Y. Ho and S. Wookey, "The real-world-weight cross-entropy loss function: Modeling the costs of mislabeling," *IEEE Access*, vol. 8, pp. 4806–4813, 2020.

[49] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[50] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016, doi: 10.1109/TGRS.2016.2543748.

[51] L. Dang, P. Pang, and J. Lee, "Depth-wise separable convolution neural network with residual connection for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 20, p. 3408, Oct. 2020.

[52] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[53] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.